

KHORRAMABAD, Iran, Islamic Republic ofWMO: **407820**Lat: **33.433N**Lon: **48.283E**Elev: **1148**StdP: **88.27**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-2.4	-13.0	1.4	9.6	-11.0	1.7	10.1	8.7	7.8	7.5	7.3	1.1	120	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	41.1	17.3	40.0	17.1	38.9	16.9	19.0	36.9	18.2	36.7	17.5	36.2	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.8	11.3	20.9	12.2	10.2	18.6	11.4	9.6	18.2	59.0	37.0	55.9	36.6	53.8	36.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.3	7.0	6.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-7.0	43.2	2.9	0.9	-9.1	43.9	-10.8	44.5	-12.5	45.0	-14.6	45.7		
			-7.9	19.6	2.8	1.5	-9.9	20.8	-11.6	21.7	-13.1	22.5	-15.1	23.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	5.0	7.2	11.1	14.9	20.8	26.8	30.1	29.7	24.6	18.8	11.3	6.9
	DBStd	9.18	2.81	3.15	3.24	3.00	2.91	2.17	2.02	1.99	2.52	2.95	3.21	2.54
	HDD10.0	392	156	87	24	2	0	0	0	0	0	0	23	101
	HDD18.3	1668	414	313	225	108	12	0	0	0	0	31	213	354
	CDD10.0	3068	1	8	59	149	334	504	624	612	438	273	61	6
	CDD18.3	1301	0	0	1	5	87	254	366	353	188	45	1	0
	CDH23.3	20481	0	0	28	135	1362	4017	5788	5543	2828	772	8	0
	CDH26.7	12741	0	0	3	23	592	2569	3914	3734	1644	259	0	0
Wind	WSAvg	2.3	2.1	2.4	2.6	2.5	2.6	2.6	2.5	2.5	2.4	2.2	1.8	1.8
Precipitation	PrecAvg	563	81	75	102	91	19	2	1	0	2	30	69	91
	PrecMax	847	131	218	221	212	60	12	3	2	16	133	292	209
	PrecMin	331	10	16	15	19	0	0	0	0	0	1	1	6
	PrecStd	128	33	48	54	50	17	3	1	1	4	35	56	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	19.5	26.8	29.2	35.2	40.8	42.9	42.1	38.9	32.9	24.5	19.9
		MCWB	7.4	9.3	11.8	14.8	15.5	17.5	17.8	17.6	16.5	14.3	11.4	8.3
	2%	DB	14.1	17.2	23.0	26.2	33.6	39.0	41.4	41.0	37.1	31.1	22.2	16.2
		MCWB	6.7	8.0	10.2	13.7	15.1	16.6	17.4	17.1	16.0	13.8	10.8	8.0
	5%	DB	12.8	15.8	20.8	24.8	31.9	38.0	40.2	40.0	35.9	29.8	20.1	14.3
		MCWB	6.5	7.5	9.7	12.9	14.5	16.3	17.2	17.0	15.3	13.3	10.2	7.5
	10%	DB	11.0	13.9	18.8	22.9	30.1	36.9	39.0	38.9	34.3	27.9	18.0	12.7
		MCWB	5.5	6.8	9.1	12.3	14.1	15.8	17.1	16.9	14.7	12.6	9.7	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	11.1	12.9	16.5	17.7	18.8	22.5	20.0	18.5	16.0	14.0	10.9
		MCDB	12.6	15.8	20.3	24.6	29.7	35.9	36.4	37.5	35.5	27.6	18.9	14.9
	2%	WB	8.3	9.5	11.9	15.1	16.7	17.7	19.5	18.7	17.1	15.0	12.7	9.5
		MCDB	11.7	14.5	20.1	23.5	28.5	35.9	36.8	37.9	33.4	26.1	17.5	13.3
	5%	WB	7.4	8.3	10.9	14.2	16.0	17.1	18.4	17.9	16.0	14.1	11.8	8.6
		MCDB	11.0	14.0	18.1	21.6	27.0	35.6	37.1	37.5	33.1	24.8	17.3	12.2
	10%	WB	6.3	7.5	9.9	13.3	15.3	16.4	17.6	17.3	15.2	13.4	11.0	7.8
		MCDB	9.9	13.1	16.8	20.3	26.3	34.8	36.9	37.1	32.7	24.2	16.1	11.5
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.1	13.5	13.8	16.4	19.4	18.3	18.9	19.7	16.6	13.1	11.8
		MCDBR	14.1	14.9	16.8	17.3	19.2	20.7	19.3	19.5	20.3	18.9	16.0	14.9
		MCWBR	7.9	7.7	7.5	7.1	6.4	7.3	6.5	6.7	7.6	7.9	7.5	7.8
	5% WB	MCDBR	11.1	12.4	14.0	14.1	16.3	19.2	17.7	18.4	18.9	15.5	11.4	10.9
		MCWBR	6.7	6.8	6.9	6.4	6.2	7.0	6.7	6.7	7.3	6.7	5.8	6.3
Clear-Sky Solar Irradiance	taub	0.349	0.401	0.470	0.572	0.547	0.488	0.548	0.501	0.424	0.462	0.394	0.350	
	taud	2.200	2.045	1.868	1.687	1.757	1.878	1.755	1.856	2.027	1.924	2.093	2.217	
	Ebn at Noon	845	832	801	738	763	807	757	786	834	755	780	818	
	Edh at Noon	114	149	192	241	227	201	226	201	161	166	126	107	
All-Sky Solar Radiation	RadAvg	2.86	3.73	4.95	5.77	7.01	8.33	7.80	7.30	6.38	4.63	3.27	2.65	
	RadStd	0.17	0.29	0.43	0.51	0.49	0.27	0.28	0.27	0.22	0.34	0.30	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	+0.62	-1.29	-1.05	N/A	N/A	N/A	N/A

Nomenclature: See separate page